IGCSE 0607 Practice Questions — E1.10

Ratio and proportion

Total Questions: 6 — **Total Marks:** 37 Marks

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1 **(a)** Write the ratio 45 minutes to 2 hours in its simplest form.

..... **[2]**

(b) A piece of wood of length 3.6 m is cut into three pieces in the ratio 2 : 3 : 7.
Calculate the length, in cm, of the longest piece.

..... cm **[3]**

2 Alex, Ben and Charlie share \$840 in the ratio 2 : 3 : 5.

(a) Calculate the amount of money Ben receives.

\$ ..... **[2]**

(b) Express Charlie's share as a fraction of the total amount, giving your answer in its simplest form.

..... **[2]**

3 A map is drawn to a scale of 1 : 250 000.

(a) The distance between two cities on the map is 16.8 cm.
Calculate the actual distance between the cities, in km.

..... km **[2]**

(b) A lake has an actual area of 56 km^2.
Calculate the area of the lake on the map, in cm^2.

..... cm^2 **[2]**

(c) On a different map of the same region, the area of the same lake is 14 cm^2.
Find the scale of this new map in the form 1 : n.

1 : ..... **[3]**

4 Three partners, Alex, Ben and Chris, share a total profit of $\$P$. The ratio of Alex's share to Ben's share is 3 : 4. The ratio of Ben's share to Chris's share is 5 : 6.

(a) Find the ratio of Alex's share to Ben's share to Chris's share in its simplest form.

..... **[2]**

(b) Chris receives \$16 200 more than Alex. Calculate the value of P.

\$ ..... **[3]**

(c) Alex invests his share of the profit in a bank account that pays compound interest at a rate of 4.2% per year.
Calculate the value of Alex's investment at the end of 5 years.
Give your answer correct to the nearest dollar.

\$ ..... **[3]**

5 A map is drawn to a scale of 1 : 250 000.

(a) The distance between two towns on the map is 8.4 cm. Calculate the actual distance between the two towns in kilometres.

..... km **[3]**

(b) A forest has an actual area of 35 square kilometres.
Calculate the area of the forest on the map in square centimetres.

..... cm^2 **[3]**

6 In a factory, the number of male workers to female workers is in the ratio 5 : 3.
The ratio of male workers who work full-time to male workers who work part-time is 3 : 2.
The ratio of female workers who work full-time to female workers who work part-time is 4 : 1.

(a) Find the ratio of the total number of full-time workers to the total number of part-time workers in the factory.
Give your answer in its simplest form.

..... **[4]**

(b) There are 42 more full-time male workers than part-time female workers.
Calculate the total number of workers in the factory.

..... **[3]**